

Cloud Gaming Platform Modernization Report (May 2026)

Scope Completed

The codebase was extracted from:

```
/home/ubuntu/Uploads/Cloudgaming-platform-main.zip
```

and modernized in:

```
Cloudgaming-platform-main/cloud_gaming_system
```

Goals completed:

- Upgraded Python dependencies to current compatible stable releases.
 - Upgraded Node.js/TypeScript/React dependencies to current stable releases.
 - Added npm lockfiles for reproducible installs.
 - Fixed compatibility/deprecation issues introduced by newer runtime/tooling versions.
 - Updated Linux service/install configuration to be safer and less brittle on modern Ubuntu.
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1) Python dependency updates (vm-host-linux/requirements.txt)

Updated from older pins to modern stable versions:

- aiortc : 1.6.0 → 1.14.0
- aiohttp : 3.9.1 → 3.13.5
- python-socketio[asyncio_client] : 5.10.0 → 5.16.1
- av : 11.0.0 → 16.1.0 (latest compatible with aiortc 1.14.0)
- numpy : 1.24.3 → 2.4.4
- opencv-python : 4.8.1.78 → 4.13.0.92
- pynput : 1.7.6 → 1.8.1
- python-xlib : kept at 0.33 (latest)
- PyYAML : 6.0.1 → 6.0.3
- python-dotenv : 1.0.0 → 1.2.2
- coloredlogs : kept at 15.0.1 (latest)

Python compatibility fixes made

- screen_capture.py
- Replaced thread use of `asyncio.get_event_loop()` with explicit capture of running loop and `call_soon_threadsafe`.
- Improved queue handling to avoid deadlocks / dropped-loop issues under modern asyncio behavior.
- Removed unnecessary stderr pipe pressure by routing ffmpeg stderr to `DEVNULL`.
- config.py

- Made config path robust and absolute by default (`vm-host-linux/config/config.yaml`).
 - Added support for `VM_HOST_CONFIG_PATH` override.
 - Explicitly loads `.env` from project base path.
 - Hardened section writes with `setdefault` -style behavior to avoid `KeyError` on missing sections.
 - `signaling_client.py`
 - Replaced bare `except:` with `except Exception:` .
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2) Node.js / npm dependency updates

Signaling server (`signaling-server/package.json`)

Updated to modern stable versions and Node baseline:

- Added engines: `node >=20`
- `express` : `^4.18.2` → `^5.2.1`
- `socket.io` : `^4.6.1` → `^4.8.3`
- `uuid` : `^9.0.0` → `^14.0.0`
- `cors` : `^2.8.5` → `^2.8.6`
- `dotenv` : `^16.0.3` → `^17.4.2`
- `winston` : `^3.8.2` → `^3.19.0`
- `helmet` : `^7.0.0` → `^8.1.0`
- `@types/express` : `^4.17.17` → `^5.0.6`
- `@types/node` : `^18.15.11` → `^25.6.0`
- `@types/cors` : `^2.8.13` → `^2.8.19`
- `typescript` : `^5.0.4` → `^6.0.3`
- `@typescript-eslint/*` : upgraded to `^8.59.2`
- `eslint` : `^8.38.0` → `^10.3.0`
- Replaced dev runtime: `ts-node-dev` → `tsx`

Web client (`web-client/package.json`)

Updated to modern stable versions and Node baseline:

- Added engines: `node >=20`
- `react` : `^18.2.0` → `^19.2.5`
- `react-dom` : `^18.2.0` → `^19.2.5`
- `socket.io-client` : `^4.6.1` → `^4.8.3`
- `@types/react` : `^18.x` → `^19.2.14`
- `@types/react-dom` : `^18.x` → `^19.2.3`
- `@vitejs/plugin-react` : `^4.2.1` → `^6.0.1`
- `typescript` : `^5.3.3` → `^6.0.3`
- `vite` : `^5.1.3` → `^8.0.10`
- `peerDependencies` moved to React 19 line

Lockfiles

- Added `signaling-server/package-lock.json`
 - Added `web-client/package-lock.json`
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3) Deprecated syntax / breaking-change fixes

JavaScript / TypeScript

- Removed deprecated WebRTC offer flags:
- `offerToReceiveAudio` / `offerToReceiveVideo`
- Updated WebRTC helper code to avoid outdated constructor wrapping patterns and tightened stats parsing for modern lib DOM typings.
- Fixed TypeScript strict-mode issues under TS 6:
- Request handler unused params renamed (`_req`).
- Session ID parameter normalization for Express 5 type changes.
- Removed unused imports.
- Added CSS module declaration for TS (`src/global.d.ts`).

Build and tooling alignment

- `web-client` build now uses library build pipeline (`build:lib`) by default.
 - Vite config split behavior:
 - `mode=lib` : builds package library output.
 - default mode: runs example app from `examples/` .
 - Added TS deprecation compatibility toggle in signaling server tsconfig for TS6 (`ignoreDeprecations`).
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4) Linux/Ubuntu config modernization

`vm-host-linux/install.sh`

- Hardened shell script (`set -euo pipefail`).
- Uses `rsync --delete` with excludes for cleaner reproducible installs.
- Handles `.env` creation from `.env.example` .
- Keeps service install and venv setup deterministic.

`vm-host-linux/vm-host.service`

- Uses venv Python path directly in `ExecStart` .
- Added optional `EnvironmentFile=-/opt/cloud-gaming-vm-host/.env` .
- Removed brittle hardcoded XAUTHORITY path tied to specific display manager/user path.

Added env templates

- `signaling-server/.env.example`
 - `vm-host-linux/.env.example`
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5) Hardcoded paths/system dependency issues addressed

Addressed:

- Removed hardcoded display-manager-specific XAUTHORITY path in systemd service.
- Replaced fragile relative config loading with absolute base-dir config resolution.

- Added env-based override path (`VM_HOST_CONFIG_PATH`) for config file location.
- Ensured install script no longer depends on missing `.env.example` file.

Still intentionally configurable defaults:

- `http://localhost:3000` defaults in configs remain as local-development defaults and are overrideable via `.env`.

Validation performed

- Signaling server: `npm install && npm run build` ✓
- Web client: `npm install && npm run build` ✓
- VM host: created venv, `pip install -r requirements.txt`, `python -m compileall src` ✓

Updated Run Instructions

Prerequisites

- Node.js 20+
- npm 10+
- Python 3.10+
- Ubuntu/Linux with X11 for VM host capture

A) Run signaling server

```
cd cloud_gaming_system/signaling-server
cp .env.example .env
npm install
npm run build
npm start
```

B) Run VM host (dev/manual)

```
cd cloud_gaming_system/vm-host-linux
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
cp .env.example .env # if needed
python3 -m src.main
```

C) Install VM host as service (Ubuntu)

```
cd cloud_gaming_system/vm-host-linux
sudo ./install.sh
sudo systemctl start vm-host
sudo systemctl enable vm-host
sudo systemctl status vm-host
```

D) Run web client example app

```
cd cloud_gaming_system/web-client
npm install
npm run dev
```

E) Build web client library package

```
cd cloud_gaming_system/web-client
npm run build
```